

MATH 1342 – Elementary Statistics – Week 1: Introduction and the Structure of Data

Aug 25 – 27 | Introduction: What Is Statistics For? • 1.1 The Structure of Data | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 7. Apply ideas of appropriate sampling techniques and experimental design to data production.	1. Describe statistics as the process of collecting, describing, and analyzing data, and judge a striking result by how likely it is to occur by chance alone. (CO1) 2. Distinguish a sample from a population and explain how a non-representative sample biases an estimate. (CO1, CO7) 3. Identify the cases and variables in a dataset presented as a table. (CO1) 4. Classify a variable as categorical or quantitative. (CO1) 5. Identify the explanatory and response variables in a research question. (CO1) 6. Compute and compare the mean and median of a small dataset and explain how an outlier affects each. (CO2)	START HERE: Welcome; Syllabus; Textbook and Wiley Course Resources; Online Tutoring; College Policies & Student Support Services	None	Participation: Syllabus – course tour and expectations	None	Online Homework: 1.1 The Structure of Data (due Sun Aug 30, 11:59 PM)	MO3, MO4, MO5
				Day-1 class data activity: collect everyone's age, then find the mean, median, mode and boxplot together; compare the class median to the U.S. median age	MO1, MO2, MO6		
		Introduction Slides: What Is Statistics For? (Day 1)	MO1, MO2, MO6				
		e-Text: 1.1 The Structure of Data	MO3, MO4, MO5				
		Interactive Lecture: 1.1 The Structure of Data	MO3, MO4, MO5	Participation: Variables – identify cases, variables and variable types	MO3, MO4, MO5		
		Slides: 1.1 The Structure of Data	MO3, MO4, MO5				

MATH 1342 – Elementary Statistics – Week 2: Sampling, Experiments and Observational Studies

Sep 1 – 3 | 1.2 Sampling from a Population • 1.3 Experiments and Observational Studies | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 7. Apply ideas of appropriate sampling techniques and experimental design to data production.	1. Distinguish a population from a sample and describe statistical inference. (CO1, CO7) 2. Identify sampling bias in a described study, classify the sampling method (random, convenience, volunteer), and judge whether the results can be generalized. (CO7) 3. Recognize bias from question wording, question context, and inaccurate responses. (CO7) 4. Classify a study as observational or experimental, and identify a plausible confounding variable. (CO7) 5. Explain how random assignment, a control group, and blinding support cause-and-effect conclusions, and distinguish random sampling (generalize) from random assignment (causation). (CO7)	e-Text: 1.2 Sampling from a Population	MO1, MO2, MO3	Participation: Variables in Statistics (Kahoot!) – Tue: review of Week 1	MO1	Online Homework: 1.2 Sampling from a Population (due Sun Sep 6, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 1.2 Sampling from a Population	MO1, MO2, MO3	Participation: Sampling – spot the bias in sample scenarios	MO2, MO3		
		Slides: 1.2 Sampling from a Population (includes sampling video)	MO1, MO2, MO3	Participation: Sampling Methods (Kahoot!) – Thu: sampling review	MO2, MO3	Online Homework: 1.3 Experiments and Observational Studies (due Sun Sep 6, 11:59 PM)	MO4, MO5
		e-Text: 1.3 Experiments and Observational Studies	MO4, MO5	Participation: Experiments and Observational Studies – classify studies; confounding; random sampling vs. random assignment	MO4, MO5		
		Interactive Lecture: 1.3 Experiments and Observational Studies	MO4, MO5				
		Slides: 1.3 Experiments and Observational Studies (includes TED-Ed placebo video)	MO4, MO5			Take-Home Quiz 1 (due Tue Sep 8, 11:59 PM) – covers Intro, 1.1–1.3	All

MATH 1342 – Elementary Statistics – Week 3: Describing Categorical and Quantitative Variables

Sep 8 – 10 | 2.1 Categorical Variables • 2.2 One Quantitative Variable: Shape and Center | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 6. Analyze data presented in two-way tables to provide information about relationships between categorical variables.	1. Construct frequency and relative frequency tables, compute a sample proportion $\hat{p}$, and distinguish the parameter p from the statistic $\hat{p}$. (CO1) 2. Compute and interpret overall and conditional proportions, and a difference in proportions, from a two-way table, choosing the correct denominator. (CO6) 3. Interpret bar charts, pie charts, side-by-side bar charts, and segmented bar charts. (CO1, CO6) 4. Describe the shape of a distribution (symmetric, left-skewed, right-skewed) from a dotplot or histogram. (CO1) 5. Compute the mean and median of a dataset and explain how skewness and outliers affect each (resistance). (CO2) 6. Interpret the standard deviation as the typical distance from the mean and apply the 95% rule to estimate the mean and standard deviation from a bell-shaped dotplot. (CO2, CO4)	e-Text: 2.1 Categorical Variables	MO1, MO2, MO3	Participation: Frequency and Relative Frequency	MO1	Online Homework: 2.1 Categorical Variables (due Sun Sep 13, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 2.1 Categorical Variables	MO1, MO2, MO3	Participation: Finding Proportions from Two-Way Tables	MO2, MO3		
		Slides: 2.1 Categorical Variables	MO1, MO2, MO3	Participation: Quantitative Variables – shape, mean vs. median	MO4, MO5		
		e-Text: 2.2 One Quantitative Variable	MO4, MO5, MO6	In-class estimation: estimate the mean and SD of a dotplot without calculating (95% rule)	MO6	Online Homework: 2.2 One Quantitative Variable (due Sun Sep 13, 11:59 PM)	MO4, MO5, MO6
		Interactive Lecture: 2.2 One Quantitative Variable	MO4, MO5, MO6				
		Slides: 2.2 One Quantitative Variable	MO4, MO5, MO6				

MATH 1342 – Elementary Statistics – Week 4: Percentiles, Boxplots and Two-Variable Relationships

Sep 15 – 17 | 2.3 One Quantitative Variable: Percentiles • 2.4 Two Variable Relationships | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
1. Interpret ideas of population versus sample, random variables, and techniques of descriptive statistics including frequency distributions, histograms, boxplots, and scatterplots. 2. Calculate and interpret measures of central tendency and dispersion, including mean, median, standard deviation, and quartiles. 4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 5. Analyze relationships between two quantitative variables using correlation and linear regression.	1. Compute the sample standard deviation of a small dataset by hand. (CO2) 2. Compute and interpret z-scores to compare values measured on different scales or in different groups. (CO2, CO4) 3. Find the five-number summary, range, and IQR; apply the 1.5×IQR rule to identify outliers; and construct or interpret a boxplot. (CO1, CO2) 4. Compare a quantitative variable across groups with side-by-side boxplots, and compute and interpret a difference in means. (CO1, CO2) 5. Describe the direction, form, and strength of the relationship in a scatterplot and estimate the correlation r. (CO5) 6. Use technology to compute r and explain how outliers, nonlinear patterns, and the absence of causation limit its interpretation. (CO5)	e-Text: 2.3 One Quantitative Variable: Using Percentiles	MO1, MO2, MO3	Participation: One Quantitative Variable (Kahoot!) – Tue: review of 2.2	MO1	Online Homework: 2.3 One Quantitative Variable: Using Percentiles (due Sun Sep 20, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 2.3 One Quantitative Variable: Using Percentiles	MO1, MO2, MO3	Participation: Using Percentiles – Your Turn: comparing heights with z-scores	MO2, MO3		
		Slides: 2.3 One Quantitative Variable: Using Percentiles	MO1, MO2, MO3	Participation: Box Plots and Outliers – five-number summary, 1.5×IQR rule	MO3	Online Homework: 2.4 Two Variable Relationships (due Sun Sep 20, 11:59 PM)	MO4, MO5, MO6
		e-Text: 2.4 Two Variable Relationships	MO4, MO5, MO6	Quick Self-Quizzes: tea vs. coffee boxplots and difference in means; NBA rebounds vs. free-throw % scatterplot; find r with and without an outlier	MO4, MO5, MO6		
		Slides: 2.4 Two Variable Relationships	MO4, MO5, MO6			Take-Home Quiz 2 (due Tue Sep 22, 11:59 PM) – covers 2.1–2.4	All

MATH 1342 – Elementary Statistics – Week 5: Linear Regression

Sep 22 – 24 | 2.5 Two Quantitative Variables: Linear Regression • Catch-Up Day (Thu Sep 24) | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
5. Analyze relationships between two quantitative variables using correlation and linear regression.	1. Interpret the slope and intercept of a regression line in context with correct units, and judge whether the intercept is meaningful. (CO5) 2. Use a regression equation to predict a response value. (CO5) 3. Compute and interpret a residual, and locate positive and negative residuals on a scatterplot. (CO5) 4. Use technology to find the least-squares regression line and explain what “least squares” means. (CO5) 5. Recognize when a regression prediction is not appropriate: extrapolation, nonlinearity, influential outliers, and causal claims. (CO5)	e-Text: 2.5 Two Quantitative Variables: Linear Regression	All	Participation: Two Variable Relationships – Tue: correlation review, Your Turn: spotting residuals	MO3	Online Homework: 2.5 Two Quantitative Variables:	All
		Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 1)	MO1, MO2, MO3	Participation: Regression Line – Thu: baseball game length and used-car price problems	MO1, MO2, MO3, MO5	Linear Regression (due Sun Sep 27, 11:59 PM)	
		Interactive Lecture: Section 2.5 Two Quantitative Variables: Linear Regression (Part 2)	MO4, MO5	Example 2 (commute distance and time): find the regression line with technology and predict	MO2, MO4	Practice Test 1 (Blackboard, due Thu Oct 1, 11:59 PM; 120 min, proctoring practice)	None
		Slides: 2.5 Two Quantitative Variables: Linear Regression	All				

MATH 1342 – Elementary Statistics – Week 6: Probability Rules and Sampling Distributions

Sep 29 – Oct 1 | P.1 Probability Rules • 3.1 Sampling Distributions | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#		
6. Analyze data presented in two-way tables to provide information about relationships between categorical variables. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions.	1. Compute probabilities for equally likely outcomes and apply the complement rule. (SLO 3) 2. Apply the addition, multiplication, and conditional probability rules using a two-way table, and distinguish disjoint events from independent events. (CO6) 3. Classify a numerical summary as a parameter or a statistic and use the correct notation (μ , σ , p , ρ vs. $\bar{x}$, s , $\hat{p}$, r). (CO8) 4. Describe how a sampling distribution is built from repeated samples, what each dot represents, and its center and shape. (CO8) 5. Explain the standard error as the variability of a statistic and predict how sample size affects it. (CO8)	e-Text: P.1 Probability Rules	MO1, MO2	Participation: Probability – Rock and Roll Hall of Fame two-way table; cards with and without replacement	MO1, MO2	Online Homework: P.1 Probability Rules (due Sun Oct 4, 11:59 PM)	MO1, MO2		
		Interactive Lecture: P.1 Probability Rules	MO1, MO2						
		Slides: P.1 Probability Rules	MO1, MO2	Participation: Sampling Distributions – M&M activity: each student draws 10 M&Ms, computes $\hat{p}$ (blue), and adds it to a class dotplot; name the population, sample, and what each dot represents	MO3, MO4, MO5	Online Homework: 3.1 Sampling Distributions (due Sun Oct 4, 11:59 PM)	MO3, MO4, MO5		
		e-Text: 3.1 Sampling Distributions	MO3, MO4, MO5						
		Interactive Lecture: 3.1 Sampling Distributions	MO3, MO4, MO5						
		Slides: 3.1 Sampling Distributions	MO3, MO4, MO5						
								Take-Home Quiz 3 (due Tue Sep 29, 11:59 PM) – covers 2.5	None
								Test 1 – Fri Oct 2 – Sun Oct 4 (Blackboard, proctored, 2 hours). Covers Weeks 1–5.	None

MATH 1342 – Elementary Statistics – Week 7: Confidence Intervals and the Bootstrap

Oct 6 – 8 | 3.2 Understanding and Interpreting Confidence Intervals • 3.3 Constructing Bootstrap Confidence Intervals Using Standard Error | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results.	1. Construct a 95% confidence interval for a mean or proportion as statistic $\pm 2 \cdot \text{SE}$, given the statistic and its standard error. (CO9) 2. Interpret a confidence interval in context, explain the confidence level as the long-run capture rate of the method, and identify common misinterpretations. (CO9) 3. Describe how a bootstrap sample and a bootstrap distribution are created, and explain why the bootstrap distribution is centered at the sample statistic. (CO3) 4. Use StatKey to generate a bootstrap distribution and estimate the standard error as its standard deviation. (CO3) 5. Construct and interpret a 95% bootstrap confidence interval using the standard-error method. (CO3, CO9)	e-Text: 3.2 Understanding and Interpreting Confidence Intervals	MO1, MO2	Participation: Interpreting Confidence Intervals – which of 20 simulated CIs capture μ ; fix three misinterpretations	MO1, MO2	Online Homework: 3.2 Understanding and Interpreting Confidence Intervals (due Sun Oct 11, 11:59 PM)	MO1, MO2
		Interactive Lecture: 3.2 Understanding and Interpreting Confidence Intervals	MO1, MO2	Participation: Bootstrap a Proportion – Try It: 32 of 50 prefer Brand A; build the bootstrap distribution, read off SE, form the CI	MO3, MO4, MO5		
		Slides: 3.2 Understanding and Interpreting Confidence Intervals	MO1, MO2			Online Homework: 3.3 Constructing Bootstrap Confidence Intervals (due Sun Oct 11, 11:59 PM)	MO3, MO4, MO5
		e-Text: 3.3 Constructing Bootstrap Confidence Intervals	MO3, MO4, MO5				
		Interactive Lecture: 3.3 Constructing Bootstrap Confidence Intervals	MO3, MO4, MO5				
		Slides: 3.3 Constructing Bootstrap Confidence Intervals (embedded StatKey simulator)	MO3, MO4, MO5			Test 1 Retake – Mon Oct 5 – Sun Oct 11 (optional; higher score counts)	None
		StatKey (lock5stat.com/statkey) – Bootstrap Confidence Intervals	MO4, MO5				

MATH 1342 – Elementary Statistics – Week 8: Percentile Intervals and Introducing Hypothesis Tests

Oct 13 – 15 | 3.4 Bootstrap Confidence Intervals Using Percentiles • 4.1 Introducing Hypothesis Tests | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Determine which percentiles of a bootstrap distribution bound a 90%, 95%, or 99% confidence interval, and construct the interval with StatKey. (CO3, CO9) 2. Predict how changing the confidence level or the sample size affects the width of an interval. (CO9) 3. Judge whether a bootstrap confidence interval is trustworthy from the shape of the bootstrap distribution and whether the sample is representative. (CO3) 4. Identify the parameter of interest and write null and alternative hypotheses from a research question. (CO10) 5. Classify a test as right-tailed, left-tailed, or two-tailed, and explain the logic of assuming H_0 and asking whether the data are surprising. (CO10)	e-Text: 3.4 Bootstrap Confidence Intervals Using Percentiles	MO1, MO2, MO3	Participation: Percentile Confidence Intervals – StatKey: 90/95/99% intervals for the same data; compare SE and percentile methods	MO1, MO2, MO3	Online Homework: 3.4 Bootstrap Confidence Intervals Using Percentiles (due Sun Oct 18, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 3.4 Bootstrap Confidence Intervals Using Percentiles	MO1, MO2, MO3	Participation: Writing Hypotheses – Zener-card ESP practice (26 of 50): define the parameter, state H_0 and H_a , choose the tail	MO4, MO5	Online Homework: 4.1 Introducing Hypothesis Tests (due Sun Oct 18, 11:59 PM)	MO4, MO5
		Slides: 3.4 Bootstrap Confidence Intervals Using Percentiles	MO1, MO2, MO3	Think about it: is 15 of 16 correct (dolphin study) convincing?	MO5	Take-Home Quiz 4 (due Tue Oct 13, 11:59 PM) – covers P.1, 3.1–3.3	None
		e-Text: 4.1 Introducing Hypothesis Tests	MO4, MO5				
		Interactive Lecture: 4.1 Introducing Hypothesis Tests	MO4, MO5				
		Slides: 4.1 Introducing Hypothesis Tests	MO4, MO5				

MATH 1342 – Elementary Statistics – Week 9: P-values and Statistical Significance

Oct 20 – 22 | 4.2 Measuring Evidence with p-values • 4.3 Determining Statistical Significance | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Describe how a randomization distribution is generated assuming H_0 is true, and identify its center as the null value. (CO3) 2. Compare sampling, bootstrap, and randomization distributions by their center and purpose. (CO3, CO8) 3. Compute a p-value from a randomization distribution as a proportion of simulated statistics, choosing the correct tail(s) from H_a . (CO3, CO10) 4. Interpret a p-value in context and relate its size to the strength of evidence against H_0 . (CO10) 5. Compare the p-value to α , decide whether to reject H_0 , write a two-step conclusion in context, and avoid saying “accept H_0 ”. (CO10)	e-Text: 4.2 Measuring Evidence with P-Values	MO1, MO2, MO3, MO4	Participation: Randomization Distributions – StatKey: dolphin study (16 flips) and tea vs. coffee re-randomization; count how many simulated statistics are as extreme as observed	MO1, MO2, MO3	Online Homework: 4.2 Measuring Evidence with P-Values (due Sun Oct 25, 11:59 PM)	MO1, MO2, MO3, MO4
		Interactive Lecture: 4.2 Measuring Evidence with P-Values	MO1, MO2, MO3, MO4			Online Homework: 4.3 Determining Statistical Significance (due Sun Oct 25, 11:59 PM)	MO4, MO5
		Slides: 4.2 Measuring Evidence with P-Values	MO1, MO2, MO3, MO4	Participation: P-values and Conclusions – resveratrol examples ($p = 0.04, 0.12, 0.0003$): strength of evidence, decision at α , conclusion in context	MO4, MO5	Take-Home Quiz 5 (due Tue Oct 27, 11:59 PM) – covers 3.4, 4.1–4.3	All
		e-Text: 4.3 Determining Statistical Significance	MO4, MO5				
		Interactive Lecture: 4.3 Determining Statistical Significance	MO4, MO5				
		Slides: 4.3 Determining Statistical Significance (includes “What is a p-value?” video)	MO4, MO5				
		StatKey – Randomization Hypothesis Tests	MO1, MO3				

MATH 1342 – Elementary Statistics – Week 10: A Closer Look at Testing and Making Connections

Oct 27 – 29 | 4.4 A Closer Look at Testing • 4.5 Making Connections | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Describe Type I and Type II errors in the context of a study, relate α to the probability of a Type I error, and explain the trade-off between the two errors and the effect of sample size. (CO10) 2. Explain how multiple testing and publication bias produce false positives, and distinguish statistical significance from practical significance. (CO10) 3. Describe how to generate randomization samples for a given study design (coin flips, re-randomizing groups, shifting data then bootstrapping). (CO3) 4. Compare bootstrap and randomization distributions by their center and use, identifying each center for given data. (CO3) 5. Use a 95% confidence interval to reach a two-tailed test conclusion at $\alpha = 0.05$, and decide whether a confidence interval or a hypothesis test fits a research question. (CO9, CO10)	e-Text: 4.4 A Closer Look at Testing	MO1, MO2	Participation: Errors in Testing – battery-life example (a)–(d): describe each error in context; courtroom analogy matching; vitamin E multiple-testing questions	MO1, MO2	Online Homework: 4.4 A Closer Look at Testing (a) (due Sun Nov 1, 11:59 PM)	MO1, MO2
		Interactive Lecture: 4.4 A Closer Look at Testing	MO1, MO2			Online Homework: 4.5 Making Connections (due Sun Nov 1, 11:59 PM)	MO3, MO4, MO5
		Slides: 4.4 A Closer Look at Testing (includes Khan Academy Type I/II video)	MO1, MO2				
		e-Text: 4.5 Making Connections	MO3, MO4, MO5	Participation: Making Connections – body temperature: test 98.6 using the CI (98.05, 98.47); sleep study (28/50): how to generate randomization samples and where each distribution is centered	MO3, MO4, MO5	Practice Test 2 (Blackboard, by Sun Nov 1)	None
		Interactive Lecture: 4.5 Making Connections	MO3, MO4, MO5			Test 2 – Fri Oct 30 – Sun Nov 1 (Blackboard, proctored, 2 hours). Covers Weeks 6–9.	None
		Slides: 4.5 Making Connections	MO3, MO4, MO5			Take-Home Quiz 6 (due Tue Nov 3, 11:59 PM) – covers 4.4, 4.5	All

MATH 1342 – Elementary Statistics – Week 11: Normal Distributions and Inference for a Proportion

Nov 3 – 5 | 5.1 Hypothesis Testing Using Normal Distributions • 6.1 Inference for a Proportion (Distribution, CI, Hypothesis Test) | Fall 2026 · Section 036 · TTh
6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Explain when the Central Limit Theorem justifies replacing a simulated randomization distribution with a normal distribution $N(\text{null value}, SE)$. (CO4, CO8) 2. Compute a standardized z statistic and find the p-value as a right-tail, left-tail, or two-tail area of the standard normal distribution. (CO4, CO10) 3. Check the large-sample condition (at least 10 in each category) for inference about one proportion. (CO8) 4. Construct and interpret a confidence interval for a population proportion with z^* for 90%, 95%, or 99% confidence. (CO4, CO9) 5. Carry out a one-proportion z test and conclude in context, and explain why the CI standard error uses $\hat{p}$ while the test uses p_0. (CO4, CO10)	e-Text: 5.1 Hypothesis Tests Using Normal Distributions	MO1, MO2	Participation: Normal Distributions and z – malaria/mosquito example: compute z, shade the tail in StatKey, find the p-value	MO1, MO2	Online Homework: 5.1 Hypothesis Tests Using Normal Distributions (due Sun Nov 8, 11:59 PM)	MO1, MO2
		Interactive Lecture: 5.1 Hypothesis Tests Using Normal Distributions	MO1, MO2				
		Slides: 5.1 Hypothesis Tests Using Normal Distributions	MO1, MO2	Participation: Inference for a Proportion – 4-step CI and 4-step test for a proportion (voting example); compare the two SEs	MO3, MO4, MO5	Online Homework: 6.1 Inference for a Proportion (due Sun Nov 8, 11:59 PM)	MO3, MO4, MO5
		e-Text: 6.1 Inference for a Proportion (6.1-D Distribution, 6.1-CI Confidence Interval, 6.1-HT Hypothesis Test)	MO3, MO4, MO5				
		Interactive Lecture: 6.1 Inference for a Proportion	MO3, MO4, MO5			Test 2 Retake – Mon Nov 2 – Sun Nov 8 (optional; higher score counts)	None
		Slides: 6.1 Inference for a Proportion	MO3, MO4, MO5				
		StatKey – Theoretical Distributions (Normal)	MO2, MO4				

MATH 1342 – Elementary Statistics – Week 12: Inference for Two Proportions and for a Mean

Nov 10 – 12 | 6.3 Inference for a Difference in Proportions (Distribution, CI, Hypothesis Test) • 6.2 Inference for a Mean (Distribution, CI, Hypothesis Test) | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Classify a study as calling for one-proportion or difference-in-proportions inference, and check the conditions. (CO8) 2. Construct and interpret a confidence interval for $p_1 - p_2$, including what its sign means and whether it contains 0. (CO9) 3. Compute the pooled proportion and carry out a two-proportion z test from summary data or a two-way table. (CO10) 4. Explain why inference for a mean uses the t distribution with $df = n - 1$, and check the conditions for t procedures. (CO4) 5. Construct and interpret a t confidence interval for a population mean, and carry out a one-sample t test. (CO4, CO9, CO10)	e-Text: 6.3 Inference for a Difference in Proportions (6.3-D, 6.3-CI, 6.3-HT)	MO1, MO2, MO3	Participation: Difference in Proportions – Try It: two-proportion randomization (18/30 vs. 12/30); treatment vs. placebo two-way table practice; compare simulation with the formula	MO1, MO2, MO3	Online Homework: 6.3 Inference for a Difference in Proportions (due Sun Nov 15, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 6.3 Inference for a Difference in Proportions	MO1, MO2, MO3				
		Slides: 6.3 Inference for a Difference in Proportions (embedded StatKey two-proportion randomization)	MO1, MO2, MO3	Participation: Inference for a Mean – gribbles CI and Chips Ahoy test worked in groups	MO4, MO5	Online Homework: 6.2 Inference for a Mean (due Sun Nov 15, 11:59 PM)	MO4, MO5
		e-Text: 6.2 Inference for a Mean (6.2-D, 6.2-CI, 6.2-HT)	MO4, MO5				
		Interactive Lecture: 6.2 Inference for a Mean	MO4, MO5				Take-Home Quiz 7 (due Tue Nov 17, 11:59 PM) – covers 5.1, 6.1–6.3
		Slides: 6.2 Inference for a Mean	MO4, MO5				
		StatKey – Theoretical Distributions (t)	MO4, MO5				

MATH 1342 – Elementary Statistics – Week 13: Inference for Two Means and Paired Data

Nov 17 – 19 | 6.4 Inference for a Difference in Means (Distribution, CI, Hypothesis Test) • 6.5 Paired Difference in Means | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 7. Apply ideas of appropriate sampling techniques and experimental design to data production. 8. Use the sampling distributions of sample proportions and sample means to answer appropriate questions. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations.	1. Compute the standard error and conservative degrees of freedom for a difference in two independent means, and check the conditions. (CO4, CO8) 2. Construct and interpret a confidence interval for $\mu_1 - \mu_2$, explaining how the order of the groups affects the sign. (CO9) 3. Carry out a two-sample t test, choosing the tail from the research question. (CO10) 4. Classify a study design as paired or independent samples by checking for a link between observations. (CO7, CO10) 5. Define the difference variable d, carry out a paired t test and paired t confidence interval, and explain why pairing reduces variability. (CO9, CO10)	e-Text: 6.4 Inference for a Difference in Means (6.4-D, 6.4-CI, 6.4-HT)	MO1, MO2, MO3	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test	MO1, MO2, MO3	Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 6.4 Inference for a Difference in Means	MO1, MO2, MO3	Participation: Paired or Independent? – class discussion of eight study scenarios (a)–(h); define d and the order of subtraction	MO4, MO5	Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM)	MO4, MO5
		Slides: 6.4 Inference for a Difference in Means	MO1, MO2, MO3				
		e-Text: 6.5 Paired Difference in Means	MO4, MO5			Test 3 – Fri Nov 20 – Sun Nov 22 (Blackboard, proctored, 2 hours). Covers Weeks 10–13.	All
		Interactive Lecture: 6.5 Paired Difference in Means	MO4, MO5				
		Slides: 6.5 Paired Difference in Means	MO4, MO5				

MATH 1342 – Elementary Statistics – Week 14: Chi-Square Test for Association

Nov 24 – 26 | 7.2 Testing for an Association Between Two Categorical Variables • Thanksgiving Holiday – No Class (Thu Nov 26) | Fall 2026 • Section 036 • TTh
6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 6. Analyze data presented in two-way tables to provide information about relationships between categorical variables. 11. Demonstrate skills in hypothesis testing using chi-squared tests.	1. State the hypotheses for a chi-square test of association from a two-way table. (CO6, CO11) 2. Compute expected counts and check that all expected counts are at least 5. (CO11) 3. Compute the chi-square statistic and $df = (r - 1)(c - 1)$, and interpret the right-tail p-value. (CO4, CO11) 4. Relate the chi-square test on a 2x2 table to the two-proportion z test, and explain that a significant result shows association but not direction or causation. (CO11)	e-Text: 7.2 Testing for an Association Between Two Categorical Variables	All	Participation: Chi-Square Test – faculty rank × gender: compute expected counts, cell contributions, χ^2 and df; decide and interpret	MO1, MO2, MO3, MO4	Online Homework: 7.2 Testing for an Association Between Two Categorical Variables (due Sun Nov 29, 11:59 PM)	All
		Interactive Lecture: 7.2 Testing for an Association Between Two Categorical Variables	All				
		Slides: 7.2 Testing for an Association Between Two Categorical Variables	All			Take-Home Quiz 8 (due Tue Nov 24, 11:59 PM) – covers 6.4, 6.5	None
		StatKey – Theoretical Distributions (χ^2)	MO3			Test 3 Retake – Mon Nov 23 – Sun Nov 29 (optional; higher score counts)	None

MATH 1342 – Elementary Statistics – Week 15: ANOVA and Inference for Slope and Correlation

Dec 1 – 3 | 8.1 Analysis of Variance • 9.1 Inference for Slope and Correlation | Fall 2026 • Section 036 • TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#
4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 5. Analyze relationships between two quantitative variables using correlation and linear regression. 12. Demonstrate skills in inference for regression and ANOVA techniques.	1. Identify when ANOVA is appropriate and state H_0 and H_a in symbols and words. (CO12) 2. Explain what SSG and SSE measure and how the ratio of between-group to within-group variability gives evidence against H_0 . (CO12) 3. Complete an ANOVA table (df, MS, F) from given sums of squares, interpret the F statistic and p-value, and check the conditions. (CO4, CO12) 4. State the simple linear model, distinguish the population slope β_1 and correlation ρ from the sample estimates b_1 and r , and interpret the slope in context. (CO5, CO12) 5. Use regression output to test $H_0: \beta_1 = 0$ (equivalently $\rho = 0$) with a t statistic on $n - 2$ degrees of freedom, state the conclusion in context, and explain why the slope and correlation tests give the same p-value. (CO12) 6. Construct and interpret a confidence interval for the slope, interpret R^2 as the proportion of variability explained, and check the conditions with a residual plot. (CO12)	e-Text: 8.1 Analysis of Variance	MO1, MO2, MO3	Participation: ANOVA – cuckoo-egg example: build the ANOVA table from given SS, find F, decide	MO1, MO2, MO3	Online Homework: 8.1 Analysis of Variance (due Sun Dec 6, 11:59 PM)	MO1, MO2, MO3
		Interactive Lecture: 8.1 Analysis of Variance	MO1, MO2, MO3				
		Slides: 8.1 Analysis of Variance	MO1, MO2, MO3	Participation: Inference for Slope and Correlation – SleepStudy: depression score vs. classes missed (test for correlation and slope, CI for slope, R^2); self-quiz: drinks per week vs. classes missed	MO4, MO5, MO6	Online Homework: 9.1 Inference for Slope and Correlation (due Sun Dec 6, 11:59 PM)	MO4, MO5, MO6
		e-Text: 9.1 Inference for Slope and Correlation	MO4, MO5, MO6				
		Interactive Lecture: 9.1 Inference for Slope and Correlation	MO4, MO5, MO6				
		Slides: 9.1 Inference for Slope and Correlation	MO4, MO5, MO6				
		StatKey – Theoretical Distributions (F)	MO3				

MATH 1342 – Elementary Statistics – Week 16: Putting It All Together and Final Exam

Dec 8 – 10 | Review: Putting It All Together – From Sampling to Hypothesis Testing (Tue Dec 8) • Final Exam – Thu Dec 10, online (Blackboard, proctored) | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Course-level Objectives	Module-level Objectives	Learning Materials	#	Activities	#	Assessments	#	
3. Find and use empirical probabilities in bootstrap distributions to find confidence intervals and in randomization distributions to test hypotheses. 4. Find and use theoretical probabilities from normal, t, chi-squared and F distributions to form confidence intervals and test hypotheses. Apply the 95% rule to normal and to approximately normal distributions. 7. Apply ideas of appropriate sampling techniques and experimental design to data production. 9. Estimate single means, difference of two means, single proportions and difference of two proportions using confidence intervals. Interpret the results. 10. Demonstrate skills in hypothesis testing for means and proportions, for single populations and comparison of two populations. 11. Demonstrate skills in hypothesis testing using chi-squared tests. 12. Demonstrate skills in inference for regression and ANOVA techniques.	1. Choose the correct procedure among one- and two-proportion z, one- and two-sample t, paired t, chi-square, ANOVA, and regression, and carry out a complete six-step hypothesis test. (CO9, CO10, CO11, CO12) 2. Carry out and interpret a complete confidence interval or hypothesis test, using simulation or theoretical distributions as appropriate. (CO3, CO4, CO9, CO10) 3. Interpret results in context, including what the study design allows the conclusion to claim about generalization and causation. (CO7)	Review Slides: Putting It All Together (with answer key and no-answers versions)	MO1, MO2, MO3	Participation: One Complete Test with a Partner – 10 problems, one per pair (narrator/writer roles), 5-minute presentations; extension: build the matching 95% CI	MO1, MO2, MO3	Final Exam – Thu Dec 10 (Blackboard, proctored, 2 hours, cumulative; no retake)	MO1, MO2, MO3	
		All Section Slides and Interactive Lectures (Weeks 1–15)	MO1, MO2, MO3					
		StatKey	MO2					